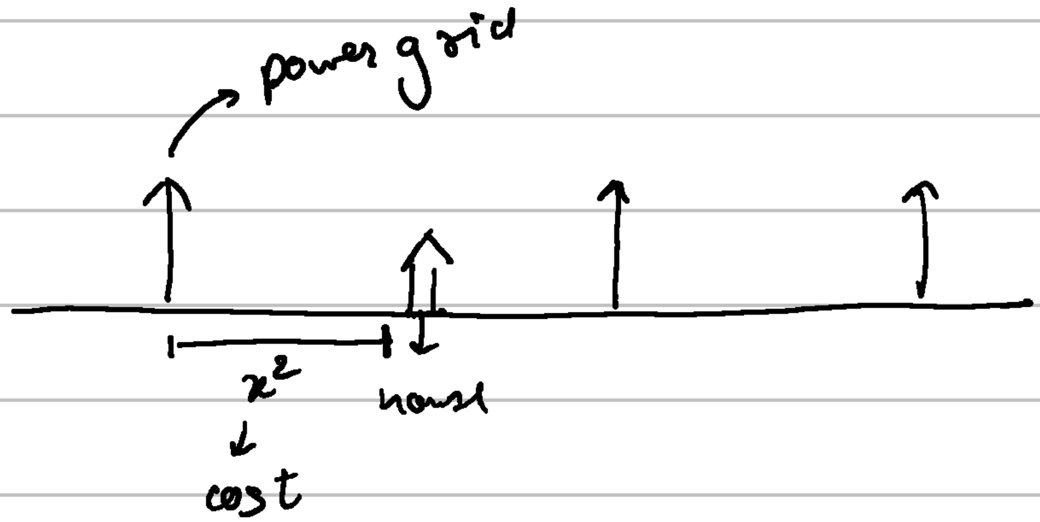
Lesson 2 - Optimization

⇒ Derivates can be used to find maxima & minima.

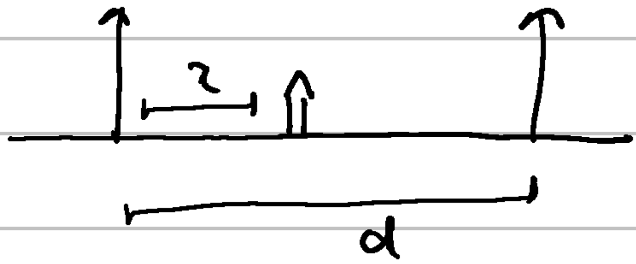
Powerline problem



⇒ Find where to put house for min cost.

① One power line → Just put it on the power line.

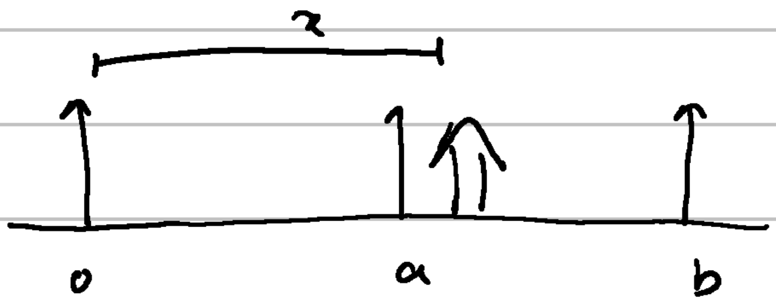
② 2 power lines



cost $= x^2 + (d-x)^2$

$$f'(x) = 2x - 2(d-x) = 0$$
$$x = d - x$$
$$2x = d$$
$$\boxed{x = \frac{d}{2}}$$ → at the center

③ 3 power lines



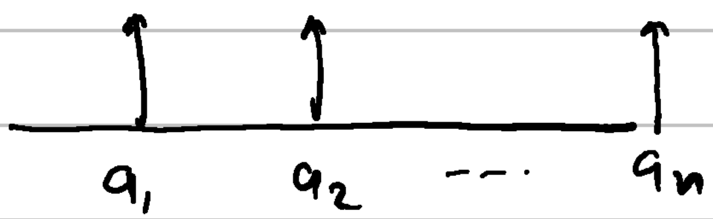
$$f(x) = x^2 + (a-x)^2 + (b-x)^2$$

$$f'(x) = \cancel{2}x - \cancel{2}(a-x) - \cancel{2}(b-x) = 0$$

$$3x = a+b$$

$$\boxed{x = \frac{a+b}{3}}$$

④ For n power lines



$$x = \frac{\Sigma a_i}{n} = \text{optimum place}$$

Optimization of log-loss

Q. Coin toss 7 heads then 3 tails
which coin is best.

1. H : 70% T : 30% ✓
2. H : 50 T : 50.
3. H : 30 T : 70

a) $\left(\frac{7}{10}\right)^7 \times \left(\frac{3}{10}\right)^3 = \frac{7^7 \cdot 3^3}{10^{10}}$ >

b) $\left(\frac{5}{10}\right)^{10}$ $\frac{5^{10}}{10^{10}}$

c) $\left(\frac{3}{10}\right)^7 \times \left(\frac{7}{10}\right)^3$ $\frac{3^7 \cdot 7^3}{10^{10}}$

Now, what is the best possible coin that we can create to maximize our probability.

$\Rightarrow$ lets say $H: x$ $\quad T: 1-x$

Probability of event $\Rightarrow$ $x^7 \cdot (1-x)^3$

$f(x) = x^7 (1-x)^3$

$f'(x) = 7 \cdot x^6 \cdot (1-x)^3 - 3 \cdot x^7 (1-x)^2$

$$\underbrace{x^6 \cdot (1-x)^2}_{\substack{0 \quad x=1 \\ \text{other extrema}}} \left(\underbrace{7 \cdot (1-x) - 3x}\right) = 0$$

$7 - 7x - 3x = 0$

$\boxed{x = \frac{7}{10}}$

So, 70 %. heads is the best coin

Trick

if $f(x)$ is maximal then so is $\log(f(x))$

$$\Rightarrow \quad f(x) = x^7 (1-x)^3$$

$$g(x) = \ln(f(x)) = 7 \ln x + 3 \cdot \ln(1-x)$$

$$g'(x) = \frac{7}{x} - \frac{3}{1-x} = 0$$

$$\boxed{x = 0.7}$$

$G(x) \rightarrow$ logarithm of probability function.

$-G(p)$ is a very common loss functions called log-loss.

⇒ The reason we take $-G(p)$ is because $G(p)$ is usually negative when $p \in [0,1]$.

Relationship with ML

① The above problem is essentially fitting the $\boxed{\text{coin}}$ to the given $\boxed{\text{dataset}}$ in a sense.

② We take logarithm because derivates of sum is easy & product is hard

③ Product of small number is very small.